

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280515

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Lunsman; Harvey John et al.

COOLING INTERFACE FOR A PLUGGABLE MODULE

Abstract

A cage for receiving a pluggable module comprises a plurality of plates including a lower plate, an upper plate, a left side plate disposed between the lower plate and the upper plate, and a right side plate disposed between the lower plate and the upper plate opposite the left side plate. A bay is defined by the plurality of plate and is sized to receive a pluggable module. The cage further includes one or more cooling interface modules. Each cooling interface module of the one or more cooling interface modules comprises a gap pad and a protective cover and is disposed such that the gap pad is adjacent to a given plate of the plurality of plates and the protective cover is adjacent to the bay.

Inventors: Lunsman; Harvey John (Chippewa Falls, WI), Scott; Michael Dustin (Chippewa Falls, WI), Ferrer; Ernesto J. (Aguadilla, PR)

Applicant: HEWLETT PACKARD ENTERPRISE DEVELOPMENT LP (Spring, TX)

Family ID: 91853554

Appl. No.: 18/591149

Filed: February 29, 2024

Publication Classification

Int. Cl.: H05K7/20 (20060101)

U.S. Cl.:

CPC H05K7/2039 (20130101);

Background/Summary

[0001] Computer systems generate heat during operation and, if the heat is not dissipated or cooling is not provided, damage can occur to various components within the system. Cooling systems come in a variety of forms, but all cooling systems help provide temperature control to allow the computer system, and the constituent components thereof, to continue functioning without damage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings and related description of the figures are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more nonlimiting aspects and implementations of the present teachings and together with the description explain certain principles and operation. In the drawings:

[0003] FIG. 1 is a diagram of an example receptacle for a pluggable module consistent with the present disclosure.

[0004] FIG. 2 is a diagram of an example information processing system including a receptacle for a pluggable module consistent with the present disclosure.

[0005] FIG. 3 is a perspective view of an example receptacle for a pluggable module consistent with the present disclosure.

[0006] FIG. 4 is an exploded perspective view of the receptacle of FIG. 3.

[0007] FIG. 5 is a front view of the receptacle of FIG. 3.

[0008] FIG. 6 is a cross-sectional view of the receptacle of FIG. 3, with section taken along line 6-6.

[0009] FIG. 7 is a cross-sectional view of the receptacle of FIG. 3 with section taken along line 7-7.

[0010] FIG. 8A is a perspective view of an example system comprising a receptacle and a directly coupled cold plate for a pluggable module consistent with the present disclosure.

[0011] FIG. 8B is a perspective view of the system of FIG. 8A in a state of a pluggable module being installed.

[0012] FIG. 9A is a perspective view of an example system comprising a receptacle and a remotely located cold plate for a pluggable module consistent with the present disclosure.

[0013] FIG. 9B is a perspective view of the system of FIG. 9A in a state of a pluggable module being installed.

[0014] FIG. 10 is a front view of an example system including receptacle in a state of a pluggable module being installed consistent with the present disclosure.

[0015] FIG. 11 is a process flow chart illustrating an example method for use of a cooling interface for a pluggable module consistent with the present disclosure.

DETAILED DESCRIPTION

[0016] Cooling systems are an important aspect of a computer, as without an adequate cooling system, the heat generated by the computer or server system may cause damage to components contained therein. One way of cooling is through liquid cooling, whereby heat is dissipated through absorption into a liquid coolant circulating through a liquid cooling system. Generally, the heat reaches the coolant through a liquid cooled cold plate that is thermally coupled to the component being cooled as well as to the liquid coolant. As used herein, a liquid cooled cold plate (hereinafter “cold plate”) refers to a device which receives heat from a solid body via conduction and dissipates the heat into a liquid coolant that is either in direct contact with the cold plate (e.g., flowing through an interior chamber thereof) or flows through a pipe or tube that is in contact with the cold plate.

[0017] Often, computer systems include pluggable modules (e.g., an optical transceiver) in addition to components that are more permanently installed (e.g., a motherboard). The pluggable modules are designed to be easily insertable and removable into a receptacle of the computing system, usually in a hot pluggable manner. The receptacle may include a cage which defines an opening, referred to herein as a bay, in which the pluggable module is received. The pluggable modules, like other components of the system, generate heat and need to be cooled. However, although cooling pluggable modules in an air-cooled system is relatively straightforward, in a computing system which is cooled primarily or exclusively by liquid cooling, it can be a challenge to cool pluggable modules due to the removable/pluggable nature of the modules. To transfer heat from the pluggable modules into the liquid coolant, a thermal interface needs to be established between the two. But, because the pluggable modules are designed to be easily insertable and removable, the approaches which are often used for cooling other components may not be suitable for the pluggable module. For example, it is typical to attach a cold plate to a central processing unit (CPU) to provide liquid cooling thereto, but attaching such a cold plate to a pluggable module would interfere with the ability to easily insert or remove the pluggable module.

[0018] One approach to providing liquid cooling to pluggable modules is to arrange a cold plate (or a thermal transfer device coupled to the cold plate) in the receptacle that receives the pluggable module such that the cold plate or thermal transfer device makes contact with the pluggable module as the pluggable module is inserted into the receptacle. However, ensuring a sufficiently high rate of heat transfer with such a thermal interface can be challenging. In particular, the surfaces which are to contact one another are generally not perfectly smooth and may not be perfectly aligned, which can lessen heat transfer rates. Surface smoothness and alignment could be improved, for example by specifying stricter tolerances, but doing so may increase costs and may also make inserting/removing modules more difficult.

[0019] One approach to increasing the heat transfer rate in spite of the surface roughness and misalignment is to add a thermal interface material (TIM) between the pluggable module and the cold plate or thermal transfer device. As used herein, a TIM refers to a relatively thin, thermally conductive, and compliant material placed between two devices at their thermal interface to improve heat transfer rates. The TIM can fill in the gaps between the mating surfaces (caused by, e.g., surface roughness or misalignment), thus improving heat transfer. However, one problem with this approach is that as a pluggable module is inserted or removed into a receptacle, sliding contact between the pluggable module and the TIM may occur and can result in damage to and/or removal of the TIM. For example, if the TIM is a gap pad, the sliding contact may tear or rub off the gap pad. Or, if the TIM is a thermal grease or paste, the sliding contact may wipe off the TIM. Thus, the damaged or removed TIM may not be effective at improving heat transfer. Furthermore, even if the TIM remains effective for one, or a few, insertions/removals, it may wear away eventually and thus may need to be reapplied periodically, which increases costs and complexity. In addition, some thermal greases or pastes can be quite messy, with the potential to spill out during insertion or to leave a residue on the removed pluggable module after removal. Thus, while the use of a TIM can improve thermal performance, there may be some drawbacks.

[0020] Another approach is to use a dry thermal interface between the module and the cold plate thermal transfer device, with “dry thermal interface” meaning a direct-contact interface with no TIM between the two contacting surfaces. With such interfaces, a relatively high contact pressure may need to be applied between the contacting surfaces in order to compensate for the roughness and/or misalignment thereof. However, achieving high contact pressures may require the user to supply a large amount of force when inserting and/or removing the pluggable modules. Generally, the amount of force a user can comfortably supply for insertion/removal is limited. In some circumstances, the amount of contact pressure needed to achieve a desired high heat transfer rate may require insertion/removal forces which exceed the limits of what the user can comfortably supply. Thus, in some systems, insertion/removal forces may be limited to a level deemed

acceptable to the average user, but as a result, the pluggable modules will have a limited amount of contact pressure with the cold plate or thermal transfer device and a correspondingly lower rate of heat transfer. In other words, to keep the module easily insertable and removable, heat transfer rates may be sacrificed. Thus, with a dry thermal interface, it is difficult to obtain both good thermal performance and easy insertion/removal.

[0021] The present disclosure addresses these issues by providing a receptacle which receives the pluggable module with one or more cooling interface modules which are configured to provide a dry thermal interface with the pluggable module capable of achieving both high heat transfer rates and low insertion/removal forces. Each cooling interface module comprises a thermal gap pad with a thin protective cover attached thereto, with the protective cover being formed from a thermally conductive material which is relatively robust (e.g., strong, hard, ductile) such as a thin metal sheet. Moreover, each cooling interface module is movably coupled to a cage of the receptacle (the cage defining the bay into which the module is inserted) such that, as the pluggable module is received into the bay, the pluggable module directly contacts and moves the protective cover, which compresses the thermal gap pad between the protective cover and the cage. Elastic restoring forces in the compressed thermal gap pad push the protective cover against the pluggable module, which conforms the protective cover to the pluggable module and alleviates any initial misalignment between the surfaces. This ensures a relatively good thermal interface, which allows heat to be transferred at a high rate from the pluggable module into the liquid cooling loop via a path comprising the protective cover, the thermal gap pad, and the cage wall. Moreover, the contact pressure between the protective cover and the pluggable modules is relatively moderate, thus avoiding the need for high insertion or removal forces. In addition, because the protective cover is what contacts the pluggable module during insertion and removal, the protective cover prevents the thermal gap pad from being damaged or removed by insertion and/or removal of the pluggable module.

[0022] In some examples, the receptacle of the present disclosure comprises a connector cage with three cooling interface modules, each comprising metal-lined gap pads, integrated with the top and side plates of the connector cage. The gap pads are sandwiched between the connector cage and the thin metal plate which lines the gap pad.

[0023] When a pluggable module is inserted into the receptacle, it slides and pushes against the thin metal plate. Because the gap pad has some “springiness” and can be compressed, the pluggable module pushing against the thin metal plate also compresses the gap pad. However, the spring force created by insertion of the pluggable module causes the gap pad to want to return to its original, uncompressed state. Thus, once the pluggable module is inserted, the gap pad pushes the thin metal plate against the pluggable module, causing the thin metal plate to conform to the pluggable module. This interference fit is what creates thermal coupling between the pluggable module and the cooling interface.

[0024] The connector cage of the receptacle is further thermally coupled to a cold plate of a cooling system. In some examples, the connector cage is directly coupled to the cold plate, i.e., the cold plate is located directly on top of the connector cage. In other examples, the cold plate is remotely located; in these examples, the heat generated by the pluggable module is conducted to the cold plate via thermal transfer devices, such as heat pipes. Regardless of the location of the cold plate, the thermal coupling of the connector cage to the cold plate allows the connector cage, and thus the pluggable module, to be integrated into the existing cooling system for the computing system.

[0025] FIG. 1 is a schematic diagram of an example receptacle 10 for receiving a pluggable module consistent with the present disclosure. It should be understood that FIG. 1 is not intended to illustrate specific shapes, dimensions, or other structural details accurately or to scale, and that implementations of the receptacle 10 may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated. In FIG. 1, physical connections (e.g., physical attachment and/or support) are indicated conceptually by dashed lines.

[0026] As shown in FIG. 1, the receptacle **10** comprises a cage **11** and one or more cooling interface modules **22**. The receptacle **10** may be installed in an information processing system (e.g., a computing system, networking device, etc.) and when so installed may provide an interface to receive a pluggable module and electrically connect a pluggable module to other components of the information processing system, such as a system board.

[0027] The cage **11** is formed by a plurality of plates, namely, a lower plate **12**, an upper plate **14**, a left side plate **18**, and a right side plate **16**, with both the left side plate **18** and the right side plate **16** being disposed between the lower plate **12** and the upper plate **14**. The plates **12**, **14**, **16**, and **18** comprise generally planar structures that form walls of the cage **11**, and are arranged such that they are all parallel to a first direction, with lower plate **12** and upper plate **14** being parallel to each other and with right side plate **16** and left side plate **18** being parallel to each other and perpendicular to lower plate **12** and upper plate **14**. In other words, the plates **12**, **14**, **16**, and **18** are coupled together to form a box-like shape (i.e., a hollow rectangular prism). In some examples, the plates **12**, **14**, **16**, and **18** are formed from one or more pieces of sheet metal which has been machined (e.g., bent, stamped, etc.) to form the shape of the cage **11**. In some examples, some or all of the plates **12**, **14**, **16**, and **18** are integrally connected to (i.e., part of the same monolithic body as) others of the plates **12**, **14**, **16**, and **18**. In some examples, some or all of the plates **12**, **14**, **16**, and **18** are formed as separate parts which are joined together by mechanical fasteners, welding/brazing, or other fastening techniques.

[0028] The cage **11** also has a bay **20** defined by the plates **12**, **14**, **16**, and **18** in the empty space located between the plates **12**, **14**, **16**, and **18**. Bay **20** may be sized to receive a pluggable module, such as an optical transceiver or other pluggable module. The cage **11** has at least one opening into the bay **20** through which the pluggable module can be inserted into the bay **20**. As described previously, a pluggable module is a component of a computer system that is designed to be selectively engaged or used, in contrast to more permanently installed components. A pluggable module is designed to be easily insertable and removable into a bay, such as bay **20**.

[0029] Although not illustrated, an electrical connector may also be disposed in or immediately adjacent to the bay **20** such that, when the pluggable module is installed in the bay **20**, a connector of the pluggable module can mate with the electrical connector of the receptacle **10**. The electrical connector may be connected to other components of an information processing system, such as a system board. In some examples, the electrical connector is part of the receptacle **10**. In other examples, the electrical connector is part of the system in which the receptacle **10** may be installed.

[0030] Although four plates are shown in FIG. 1, examples are not so limited, and fewer plates may be used. For example, a cage could be formed from an upper plate and two side plates, omitting the lower plate, and some other structure separate from the cage, such as a surface to which the cage is attached, could define a boundary of the bay in lieu of the lower plate.

[0031] The cage **11** may be formed from a thermally conductive material, such as copper, a copper alloy, nickel, steel, stainless steel, aluminum, or an aluminum alloy. Good thermal conductivity allows the cage **11** to efficiently transfer heat from a pluggable module received in the bay **20** to a cooling subsystem (not illustrated) which may be thermally coupled to the cage **11**. In addition to their good thermal conductivity, the aforementioned materials may also be beneficial for use in the cage **11** because they are electrically conductive, which can help contain electromagnetic interference (EMI), and structurally robust, which can allow the cage **11** to withstand repeated insertions and removals of the pluggable module and to provide support to the pluggable module when installed therein. In some examples, the entirety of cage **11** may be formed from one material, such as copper. In other examples, a combination of materials may be used to form cage **11**. For example, in some implementations, the upper plate **14** of cage **11** may be made of copper due to copper having a high degree of thermal conductivity, such that the upper plate **14** is primarily responsible for thermal conductivity between the pluggable module and the other components of the system. Meanwhile, the left side plate **18** and the right side plate **20** may be made of a material

with a lower degree of thermal conductivity, such as aluminum or stainless steel. The lower plate **12** may also be made of the less thermally conductive material, or may be made of copper, similar to the upper plate **14**. Examples are not so limited, however, and any combination of materials for the plurality of plates may be used to achieve a desired level of thermal conductivity of the cage **11**. [0032] At least one cooling interface module **22** is coupled to the cage **11**. Although only one cooling interface module **22** is shown in FIG. **1**, examples are not so limited and more cooling interface modules **22** may be used. The cooling interface module **22** may comprise a gap pad **24** and a protective cover **26**. As used herein, a gap pad refers to a particular type of thermal interface material (TIM) that is thermally conductive and relatively compliant, meaning that the gap pad may compress and conform to the surfaces between which it is placed. The gap pad may have a compression rate of between ten and fifty percent, and preferably has a compression rate of around twenty percent.

[0033] The cooling interface module **22** may further include a protective cover **26**. As used herein, a protective cover refers to a piece of material designed to protect the gap pad when the cooling interface module **22** is in use. The protective cover may be made of a low-friction, thermally conductive material that is also both conforming and tear-resistant. The protective cover **26** is attached to one face of the gap pad **24**, for example via adhesive. In some examples, the protective cover **26** covers the entirety of the face of the gap pad **24**. In some examples, the protective cover **26** also extends beyond the edges of the gap pad **24** in one or more directions. In some examples, the protective cover **26** includes attachment features which assist in attaching the cooling interface module **22** to the receptacle **10**.

[0034] Each cooling interface module **22** is coupled to a corresponding plate of the receptacle **10**, such that, when coupled to receptacle **10**, the gap pad **24** is sandwiched between the protective cover **26** and the plate **26**. The cooling interface module **22** protrudes into the bay **20**, with the protective cover **26** facing a central region of the bay **20**. Thus, when a pluggable module is inserted into the bay **20**, the pluggable module makes contact with the cooling interface module **22**. More particularly, the pluggable module slides along the protective cover **26** during insertion and removal of the pluggable module. The gap pad **24** is thereby protected from damage caused by the pluggable module.

[0035] In some examples, each cooling interface module **22** may be coupled to the corresponding plate by a mechanical attachment feature which retains the cooling interface module on the cage **11** while also allowing for the movement of the cooling interface module **22** in response to insertion of the pluggable module. For example, the mechanical attachment feature may include a hook at one end of the cooling interface module **22** which engages with the end of the corresponding plate. In some examples, each cooling interface module **22** may be coupled to the corresponding plate by adhesion between the gap pad **24** and the corresponding plate, which may be used in addition to or in lieu of the mechanical attachment feature. In some examples, the adhesion may be provided by a natural tackiness of the gap pad **24** itself. In other examples, the adhesion may be provided by an adhesive which is added between the gap pad **24** and the plate.

[0036] During insertion of the pluggable module, the gap pad **24** is compressed by the force of the insertion (i.e., by contact with the pluggable module). This compression occurs because the free space in the bay **20** (i.e., the unoccupied space in the middle portion of the bay **20**, which is bounded by the protective covers **26** of the cooling interface module(s) **22** and by those of the plates **12**, **14**, **16**, or **18** which lack a cooling interface module (if any)) is slightly smaller in cross-section than the pluggable module, and thus the pluggable module necessarily must move the protective covers **26** in a generally radially outward direction (i.e., towards the plates) in order to make room for itself. This radially outward motion of the protective covers **26** compresses the gap pads **23**. In addition, the protective cover **26** is deformed by the pressure between the pluggable module and the compressed or deformed gap pad **24**. As a result, the protective cover **26** conforms to the shape of the surface of the pluggable module, resulting in an improved thermal interface

between the cooling interface module **22** and the pluggable module. Thus, when the pluggable module is installed into bay **20**, a thermally conductive pathway is formed from the pluggable module to the cage **11** via the cooling interface module.

[0037] As noted above, the protective cover **26** may be made from a relatively low-friction, thermally conductive, conformable, and durable (tear resistant) material. For example, the protective cover **26** may have a coefficient of static friction less than 1.5, a thermal conductivity greater than 5 w/mK, and a tensile strength greater than 10000 psi. It is noted that the conformability and thermal conductivity of the protective cover **26** may depend not only on the intrinsic qualities of the material used to form the protective cover **26**, but also on the thickness of the protective cover **26**. For example, a relatively thin protective cover **26** may have a higher conductivity and greater conformability than a relatively thicker protective cover **26** of the same material. Many metals can be made into relatively thin sheets, foils, or films while retaining good tear resistance, and also tend to have good thermal conductivity and conformability (particularly when formed into a thin sheet or film). Thus, in some examples, the protective cover may be a thin metal sheet, foils, or film, such as a sheet, foil, or film made of copper, a copper alloy, nickel, steel, stainless steel, aluminum, an aluminum alloy, or a combination thereof. In such examples, the average thickness of the protective cover **26** may be as small as 0.0008 inches (0.020 millimeters) or, depending on the material, up to 0.006 inches (0.152 millimeters).

[0038] In particular, in some examples the protective cover **26** is copper and the average thickness thereof may be around 0.003 inches (0.076 millimeters). Copper has excellent thermal conductivity, which helps improve thermal transfer through the protective cover **26**. In addition, copper is highly malleable, which allows the cover **26** to easily conform to the surface of the pluggable module when pressed against it by the restoring forces of the compressed gap pad **24**, and this conforming of the cover **26** to the pluggable module improves the thermal interface therebetween and thus further improves the rate of heat transfer.

[0039] In other examples, the protective cover **26** is stainless steel, and the average thickness thereof may be around 0.001 inches (0.025 millimeters). Stainless steel has excellent durability, which allows it to be made thinner than other materials while still resisting tearing during insertion/removal of the pluggable module. This thinness allows for high thermal conductivity and conformability of the protective cover **26**. In particular, although stainless steel may have lower intrinsic thermal conductivity than copper, the ability to make the protective cover **26** thinner can offset the lower intrinsic thermal conductivity of the steel and produce comparable (in some cases, better) overall conductivity through the thermal interface module **22**.

[0040] Although metals in general, and copper and stainless steel in particular, are good candidates for the materials of the protective cover **26**, examples are not so limited. In other examples, the protective cover may be manufactured from a thermally conductive plastic or other thermally conductive non-metallic material, such as thermally conductive tape. Regardless of the material, protective cover **26** may be disposed adjacent to gap pad **24** so as to protect gap pad **24** when a pluggable module is engaged.

[0041] In some examples, additional elements are attached to the protective cover **26** on the same side thereof as the gap pad **24**. For example, a plurality of gap pads, including the gap pad **24**, may be attached to the same side of the protective cover **26**. In some examples in which a plurality of gap pads are attached to the same protective cover, some of gap pads may have different material properties than others.

[0042] For example, in some implementations, one or more of the gap pads may have relatively high thermal conductivity but may be less springy (i.e., produce relatively lower restoring (spring) forces when compressed), whereas one or more others of the gap pads may have relatively lower thermal conductivity but be more springy (i.e., produce relatively higher restoring (spring) forces when compressed). In such implementations, the springy gap pads may be added to aid in pushing the protective cover **26** against the pluggable module to ensure conformation thereof and a good

thermal interface, whereas the high thermal conductivity gap pads may be provided to provide good heat conduction from the protective cover **26** to the cage **11**.

[0043] FIG. **2** is a diagram of an example information processing system **200** (“system **200**”) including a receptacle **210** for a pluggable module consistent with the present disclosure. The receptacle **210** is one example implementation of the receptacle **10** described above in relation to FIG. **1**, and similar reference numbers with the same last two digits (e.g., **111** and **211**) are used for components of the receptacle **210** which correspond to (e.g., are example implementations of) components of the receptacle **10** which were already described above. In FIG. **2**, physical coupling (e.g., contact, attachment, support, etc.) are indicated by dashed lines, and electrical connections are indicated by dotted lines. System **200** may be, for example, a computing system (e.g., server), networking system (e.g., switch), or other information processing system.

[0044] System **200** includes a chassis **28** and a system board **30** supported by and/or housed within the chassis **28**. System **200** further includes a receptacle **210** and a removably pluggable module **32**.

[0045] System board **30** may be a printed circuit board assembly, or PCA, which may be housed in and/or supported by chassis **28**. In some examples, system board **30** may include a printed circuit board, or PCB, and one or more electronic components mounted thereto, such as a processor, memory, or other components.

[0046] The receptacle **210** comprises a cage **211** which may be coupled to or form a part of the chassis **28**. Similar to the receptacle **10** described in relation to FIG. **1**, the receptacle **210** may include cage **211** formed by a plurality of plates; namely, a lower plate **212**, an upper plate **214**, and a plurality of side plates **216**, **218** to couple the lower plate **212** to the upper plate **214**. In some examples, a given one of the plates may be coupled with, or may be part of, the chassis **28**. For example, in FIG. **2** the lower plate **212** is shown as being coupled with the chassis **28**. In some examples, the given plate (e.g., the lower plate **212**) may be a separate plate that is attached to the chassis **28**, for example by welding, mechanical fasteners, adhesive, or any other suitable fastening mechanism. In other examples, the given plate (e.g., the lower plate **212**) may be integrally formed with the chassis **28**. Said differently, the given plate **212** be part of a wall, panel, or other structure of the chassis **28**, such that a separately attached plate is not present.

[0047] The receptacle **210** also includes one or more cooling interface modules **222**, which are disposed within and attached to the cage **211**. Cooling interface module **222** may include a gap pad **224** and a protective cover **226**. The cooling interface module **222** may be disposed along, and coupled to, a plate of the plurality of plates. FIG. **2** shows the cooling interface module **222** disposed along and coupled to the upper plate **214**; however, examples are not so limited and the cooling interface module **222** may be disposed along the lower plate **212** or one of the plurality of side plates **216**, **218**. In addition, although only one cooling interface module **222** is shown in FIG. **2**, more than one cooling interface module **222** may be used. In such examples, each cooling interface module **222** may be disposed along a plate. For example, a first cooling interface module **222** may be disposed adjacent to a first plate of the plurality of plates **212**, **214**, **216**, **218**, and a second cooling interface module **222** may be disposed adjacent to a second plate of the plurality of plates **212**, **214**, **216**, **218**. In some examples, a third cooling interface module may be further included and may be disposed adjacent to a third plate of the plurality of plates **212**, **214**, **216**, **218**.

[0048] As described with respect to FIG. **1**, the cooling interface module **222** includes a gap pad **224** and a protective cover **226**. As shown in FIG. **2**, when installed in system **200**, the gap pad **224** may be disposed adjacent to (e.g., in contact with) the plate, in this case, upper plate **214**. The gap pad **224** may have a level of tackiness inherent to the material that may help the gap pad **224** be held into place with respect to the upper plate **214**. In addition, the protective cover **226** may have an attachment mechanism, such as a hook, to couple the cooling interface module **222** to the cage **211**. This attachment mechanism is discussed further herein with respect to FIG. **5**.

[0049] The protective cover **226** may be coupled to the gap pad **224** such that protective cover **226** and gap pad **224** are movable as a single unit. For example, protective cover **226** may be affixed to

gap pad **224** by adhesive. When installed in cage **211**, protective cover **226** may be disposed opposite from the plate that the gap pad **224** is engaged with, e.g., the upper plate **214** in the example of FIG. **1**. Said differently, cooling interface module **222** may be installed within cage **211** so that the gap pad **224** is sandwiched between one of the plates (e.g., upper plate **214**) and protective cover **226**.

[0050] This configuration may be used to create a thermal interface between a removably pluggable module **32** and the cage **211** when the pluggable module **32** is inserted into the cage **211**. More specifically, the cooling interface module **32** protrudes into the bay **220** defined by the cage **211** such that, when the pluggable module **32** is inserted into the bay **220**, the pluggable module **32** makes contact with the protective cover **226**, as indicated by dashed lines in FIG. **2**. Moreover, as the pluggable module **32** is inserted, it presses the protective cover **226** towards the corresponding plate of the cage **211** (e.g., upper plate **214** in FIG. **2**), thus compressing the gap pad **224** between the protective cover **226** and the cage plate. The gap pad **224** resists compression and generates spring-like forces which push the protective cover **226** against the pluggable module **32**. This generates contact pressures which improve a conductivity of a thermal interface between the pluggable module **32** and the protective cover **226**. In particular, in some examples the protective cover **226** is sufficiently malleable that the pressure causes it to conform to a surface of the pluggable module **32**, thus producing a good thermal interface therebetween notwithstanding the lack of a thermal interface material. In this state, pluggable module **32** is in contact with protective cover, which is in contact with gap pad, which is in contact with a plate of the cage **211** (e.g., with upper plate **214** in FIG. **2**), and therefore a conductive pathway is formed from the pluggable module **32** to the cage **211**. In other words, in this state, heat flows from the removable pluggable module **32** into the protective cover **226**, from the protective cover into the gap pad **224**, and from the gap pad **224** into the cage **211**.

[0051] A cold plate **34** may be thermally coupled to the cage **211**. As described previously, a cold plate receives heat from a solid body via conduction and dissipates the heat into a liquid coolant, cooling the solid body to which the cold plate is connected. As used herein, two items are “thermally coupled” if: (1) they are in contact with one another, or (2) they are both in contact with a thermally conductive thermal transfer device (or thermally conductive chain of thermally coupled thermal transfer devices), or (3) a heat transfer coefficient between the two items is $10 \text{ W.Math.m}^{-2}\text{.Math.K}^{-1}$ or less. Notably, “contact” includes both direct contact and indirect contact in which a thermally conductive intermediary material, such as a TIM, is present. The cold plate **34** may be part of a liquid cooling loop which flows liquid coolant through the system **200** to cool components thereof. In addition to the cold plate **34**, the liquid cooling loop may include additional cold plates (not illustrated) to cool other components, such as a processor of system board **20**, as well as other liquid cooling infrastructure (not illustrated), such as coolant lines, valves, fittings, etc.

[0052] In some examples, the cold plate **34** may be directly coupled with cage **211**, such that the cold plate **34**. FIG. **3** shows an example of such direct coupling. In other examples,, the cold plate **34** may be remotely located from cage **211**. In such examples, thermal transfer devices, such as heat pipes, may be used to thermally couple cage **211** to the cold plate **34**. A thermal transfer device is any thermally conductive device configured to transfer heat from one object thermally coupled thereto across some distance to another object thermally coupled thereof. Examples of thermal transfer devices include heat pipes, vapor chambers, a bar of copper (or other conductive material), etc. FIG. **4** shows an example of such a configuration. Because cold plate **34** is thermally coupled with the cage **211** (either directly, or via thermal transfer devices), the heat transferred into the cage **211** from the removable pluggable module **32** can be transferred into the cold plate **34**, and from there into the liquid coolant which removes the heat from the system **200**. In other words, a thermal pathway is established from the pluggable module **32** to the cold plate **34** via the cooling interface module **222** and the cage **211**. This thermal path is indicated in FIG. **2** by the thick dashed lines. In

examples in which the receptacle **210** has multiple cooling interface modules **222**, multiple thermal paths similar to the aforementioned may be established to convey heat from the pluggable module **32** to the cold plate **34**, with each such thermal path going through one of the modules **222** and a corresponding plate of the cage **211** with which that module **222** is in contact.

[0053] Turning now to FIGS. **3-7**, an example receptacle **510** for receiving a pluggable module will be described. The receptacle **510** is one example implementation of the receptacles **10** and **210** described above. Some components of the receptacle **510** correspond to (e.g., are example implementations of) components of the receptacle **10** or **210**, and such corresponding components are given similar reference numbers with the same last two digits (e.g., **511**, **211**, and **11**). Although the receptacle **510** of FIG. **3** is one example implementation of the receptacles **10** and **210**, the receptacles **10** and **210** are not limited to the receptacle **510**.

[0054] As shown in FIGS. **3-7**, receptacle **510** comprises a cage **511** and three cooling interface modules **522**. The cage **511** may be formed from a lower plate **512**, an upper plate **514**, a left side plate **516**, and a right side plate **518**. In FIG. **4**, lower plate **512** is shown as offset from upper plate **514**, left side plate **516**, and right side plate **518**; however, in use, lower plate **512** is coupled thereto to form cage **511**.

[0055] More specifically, in this example, upper plate **514**, a left side plate **516**, and a right side plate **518** are integrally connected together (i.e., part of the same monolithic body). For example, a sheet of metal which may be bent into the structure comprising the plates **512**, **514**, and **516** which is shown in FIG. **4**, and then subsequently the lower plate **512**, which is a separate piece, is attached to this structure (e.g., by welding, brazing, etc.) to complete the cage **511**. To facilitate this attachment of the lower plate **512** to the other plates **512**, **514**, and **516**, flanges **513** are provided which are integrally connected to the left and right plates **516** and **518**. The flanges **513** are bent perpendicular to the left and right plates **516** and **518** so as to be parallel to the lower plate **512**, thus providing a relatively large surface area which can be connected to the lower plate **512**. In some example, to avoid having the flanges **513** stick out into the bay **520**, the lower plate **512** may be provided with recessed or thinned regions **515** which receive the flanges **513**, such that a top surface of the flanges **513** are aligned (e.g., coplanar) with the top surface of a central portion **517** of the lower plate **512**, as shown in FIGS. **5** and **6**.

[0056] In other examples, plates **512**, **514**, **516**, and **518** may all be integrally connected. In other examples, plates **512**, **514**, **516**, and **518** may each be formed separately and then joined later.

[0057] In the example of FIGS. **2-7**, the receptacle includes three cooling interface modules **522-1**, **522-2**, **522-3** (collectively, cooling interface modules **522**). Importantly, while three cooling interface modules **522** are shown, examples are not so limited and other numbers of cooling interface modules **522** may be used. As shown in FIG. **3**, each cooling interface module **522** is disposed adjacent to and is coupled with a corresponding one of the plates **512**, **514**, **516**, and **518**. Note that in the exploded view of FIG. **4**, the cooling interface modules **522** are shown external to their respective plates of cage **511**; that is, cooling interface modules **522** are shown outside cage **510**, and not in their installed positions. When installed, cooling interface module **522-1** is disposed along left side plate **516**, cooling interface module **522-2** is disposed along upper plate **514**, and cooling interface module **522-3** is disposed along right side plate **518**.

[0058] Each cooling interface module **522** includes a gap pad, such as gap pads **524-1** to **524-3** (collectively, gap pads **524**). It is to be noted that, although not shown in FIG. **4**, cooling interface module **522-1** would have a similar gap pad **524-1**, which is visible in FIG. **6**. The gap pads **524** are disposed adjacent to a protective cover, such as protective cover **526-1**, **526-2**, **526-3** (collectively, protective cover **526**). In addition, each cooling interface module **522** includes a protective cover **526** that, as discussed with respect to FIG. **1**, is designed to protect the gap pad **524**.

[0059] As shown in FIGS. **3-5** and **7**, a hook **50-1**, **50-2**, **50-3** (collectively, hook **50**) is attached to one end of the protective cover **526** of a respective cooling interface module **522**. Each hook **50** includes a ramp portion **50a** that extends at an angle away from protective cover **526**. Specifically,

each ramp portion **50a** protrudes generally forward from the protective cover **526** and away from a center of the bay **520**. Each hook **50** also includes a cutback portion **50b** which extends generally rearward from the end of the ramp portion **50a**, extending back over at least part of the ramp portion **50a**, and in some cases also over part of the protective cover **526**. The cutback portion **50b** may, in some examples, extend parallel to the protective cover **526**. In this manner, an opening is formed into which a plate of cage **511** is able to be received (between cutback portion **50b** and ramp portion **50a**), allowing cooling interface module **522** to be connected to cage **510**. More particularly, hook **50** may allow cooling interface module **522** to be connected to an edge of cage **511** in a manner that still allows the cooling interface module **522** to move relative to the cage **511**, such as by pivoting about the hook **50**. In some examples, the hook **50** is integrally connected to the protective cover **526**—for example, the hook **50** and cover **526** may be formed from a same piece of metal sheet or foil which has been bent to form the structure shown in FIGS. 3-7. In some examples, the hook **50** may be formed from a thicker material than the protective cover **526**. In other examples, the hook **50** and protective cover **526** have similar thicknesses.

[0060] In some examples, the ramp portion **50a** of hook **50** that extends upwardly and away from protective cover **526** provides a sloped or ramped surface which forms a portion of the rim or perimeter of an opening of the bay **520**. These sloped or ramped surfaces form lead-in surfaces which are arranged to contact a pluggable module as the pluggable module is being inserted and to guide the pluggable module into the bay **520**. In addition, the sloped surfaces allow for a gradual transition from a first state in which the pluggable module is not in contact with the protective covers **526** and the gap pads **524** are not compressed to a second state in which the pluggable module is sliding along the protective covers **526** with the gap pads **524** compressed.

[0061] Turning now to FIGS. 8A and 8B, FIG. 8A is an example system **800** including receptacles and a directly coupled cold plate for a pluggable module will be described. FIG. 8B is a perspective view of the system of FIG. 8A in a state of a pluggable module being installed. Two receptacles **810-1**, **810-2** (collectively receptacles **810**) are shown, although examples are not so limited and more or fewer receptacles may be present. System **800** is one example implementation of system **200** described above. Some components of the system **800** correspond to (e.g., are example implementations of) components of the system **200**, and such corresponding components are given similar reference numbers with the same last two digits (e.g., **832** and **32**). Although the system **800** of FIGS. 8A and 8B are one example implementation of the system **200**, the system **200** is not limited to system **800**.

[0062] In some examples, the system **800** may be a computing system, such as a server. That is, the system **800** may comprise a printed circuit board assembly (PCA) **828**, which is either a primary system board of the computing system or is a secondary board configured to be connected to the primary system board of the computing system. In some examples, PCA **828** is the primary system board and comprises a sockets to receive a processor and memory.

[0063] As shown in FIG. 8A, a cold plate **834** is disposed atop and coupled to receptacles **810**. Two pipes **36-1**, **36-2** (collectively, pipes **36**) extend from cold plate **834**. Pipes **36** facilitate movement of liquid coolant, allowing heat generated when a pluggable module is inserted into receptacles **810** to be dissipated. The pipes **36** and cold plate **834** are part of a liquid cooling loop which cools system **800**. The liquid cooling loop may also include a pump and other liquid cooling infrastructure (not shown) which circulate liquid coolant, as would be familiar to those of ordinary skill in the art. The pipes **36** are arranged in the liquid cooling loop such that one of the pipes **36** receives (directly or indirectly) liquid coolant from a coolant source (e.g., a heat exchanger) and supplies the liquid coolant to the cold plate **834**, and the other one of the pipes **36** returns (directly or indirectly) the liquid coolant to the heat source. In some examples, the pipes **36** are fluidically coupled to an interior chamber within cold plate **834** through which the liquid coolant circulates, with one pipe **36** supplying the liquid coolant into the chamber and the other pipe **36** receiving outflows of the liquid coolant from the chamber. In such examples, the liquid coolant directly

contacts the interior surfaces of the cold plate **834**. In other examples, the pipes **36** are portions of the same single pipe (or pipe network) which extends through the cold plate **834**, such that the liquid coolant does not directly contact the cold plate **834** but instead contacts the pipe walls which in turn contact the cold plate **834**.

[0064] FIG. **8B** shows the system **800** of FIG. **8A** in a state of the pluggable modules **832-1**, **832-2** (collectively, pluggable modules **832**) inserted into the receptacles **810**. In this state, the pluggable modules **832** contact the cooling interface modules of the receptacles **810** and therefore a thermally conductive pathways is established between the pluggable modules **832** and the liquid coolant flowing through the cold plate **834**, with the thermally conductive pathways going from pluggable module **832** to the cooling interface modules, from cooling interface modules to the cage of the receptacle **810**, from the cage of the receptacle **810** into the cold plate **834**, and from the cold plate **834** into the liquid coolant (either directly, or via walls of the pipes **36**). System **800** may also include electrical connectors (not shown) which are disposed at a rear of each receptacle **810** and electrically connected to PCA **828**, such that, in the inserted state shown in FIG. **8B**, the pluggable module **832** mate with the electrical connectors and are electrically connected to PCA **828** thereby.

[0065] FIGS. **9A** and **9B** show an example system **900** including receptacles and a remotely located cold plate for a pluggable module. FIG. **9B** is a perspective view of the system of FIG. **9A** in a state of a pluggable module being installed. Two receptacles **910-1**, **910-2** (collectively receptacles **910**) are shown, although examples are not so limited and more or fewer receptacles may be present. System **900** is one example implementation of system **200** described above. Some components of the system **900** correspond to (e.g., are example implementations of) components of the system **200**, and such corresponding components are given similar reference numbers with the same last two digits (e.g., **932** and **32**). Although the system **900** of FIGS. **9A** and **9B** are one example implementation of the system **200**, the system **200** is not limited to system **900**. The system **900** may be similar to the system **800**, except that in the system **900**, cold plate **934** is not directly in contact with the receptacles **910**, but is instead remotely located, as described in greater detail below.

[0066] Like system **800**, system **800** may be a computing system, such as a server, and comprises a PCA **928**. This PCA **928** may be either a primary system board of the computing system or a secondary board configured to be connected to the primary system board of the computing system. In some examples, PCA **928** is the primary system board and comprises a sockets to receive a processor and memory

[0067] As shown in FIGS. **9A** and **9B** a plurality of heat pipes **40-1**, **40-2**, **40-3**, **40-4** (collectively heat pipes **40**) are coupled to receptacle **910** and to cold plate **934**. The heat pipes **40** comprise a hollow interior in which is disposed a working fluid and a wicking structure which are configured to transfer heat along the longitudinal direction of the heat pipe **40** by a repeating cycle of evaporation, convection, condensation, and wicking. That is, heat is absorbed by a liquid phase portion of the working fluid at the hot end (the end which is connected to receptacle **910**) resulting in evaporation of the heated working fluid into a vapor phase, the vapor phase working fluid then moves through convection to the cool end (the end which is connected to cold plate **934**) whereupon the vapor phase working fluid condenses back into a liquid phase and in so doing releases heat into the cold plate **934**, and then the now condensed liquid phase is wicked back to the hot end to begin the cycle again. In this manner, heat is absorbed at the hot end of each heat pipe **40** and transported to the cool end thereof, thus transporting heat out of the receptacles **910** and into the cold plate **934**. In other examples, heap pipes **40** may be replaced by other types of thermal transfer devices, which may include any thermally conductive device which can transfer heat between receptacles **910** and cold plate **934**. For example, a thermal transfer device may include a sold piece of thermally conductive material, such as copper or another thermally conductive metal. Thus, the heat generated by the pluggable modules **932** inserted into receptacles **910** reaches the cold plate **934** through the thermal connection between the receptacles **910**, the heat pipes **40** (or

other thermal transfer device used in lieu of the heat pipes **40**), and the cold plate **934**. Cold plate **934** includes a plurality of pipes **936-1**, **936-2** (collectively, pipes **936**). Pipes **936** circulate liquid coolant through cold plate **934**, causing heat pipes **40** to be cooled and thus transferring the cooling to receptacle **910**. The cold plate **934** and pipes **936** may be configured similarly to the cold plate **834** and pipe **836** described above.

[0068] FIG. **10** is a front view of an example system **1000** including a receptacle **1010** in a state of a pluggable module **1032** being installed consistent with the present disclosure. System **1000** is one example implementation of system **200** described above. Some components of the system **100** correspond to (e.g., are example implementations of) components of the system **200**, and such corresponding components are given similar reference numbers with the same last two digits (e.g., **1010** and **210**). Although the system **1000** of FIG. **10** is one example implementation of the system **200**, the system **200** is not limited to system **1000**.

[0069] As shown in FIG. **10**, system **1000** comprises a receptacle **1010**, a pluggable module **1032**, a cold plate **1034**, and a base **1028**. The receptacle **1010** is an implementation example of the receptacle **10** described above and comprises a cage **1011** and one or more cooling interface modules **1022** coupled to the cage. Each cooling interface module **1022** comprises a protective cover **1026** and a gap pad **1024**. In some examples, the receptacle **1010** may have the same configuration as the receptacle **510** described above, as shown in FIG. **10** as one illustrative example. In other examples, the receptacle **1010** may have a different configuration, such as having more or fewer cooling interface modules **1022** and/or a differently shaped cage **1011**.

[0070] In some examples, the base **1028** is a PCA, such as a primary system board or secondary board of a computing system (e.g., similar to the PCA **828** or **928**). In other examples, the base **1028** is a mechanical support, such as a wall the chassis of a computing system. The cage **1011** is coupled to the base **1028**.

[0071] In the illustrated example, the cold plate **1034** is in direct contact with the cage **1011**, specifically a top plate **1014** thereof, similar to the arrangement illustrated in FIGS. **8A** and **8B**, however in other examples the system **1000** may have a remotely located cold plate **1034**, similar to the arrangement illustrated in FIGS. **9A** and **9B**, in which case heat pipes (or other thermal transfer devices) would be in contact with the cage **1011** instead of the cold plate **1034**.

[0072] As shown in FIG. **10**, the pluggable module **1032** is received in the cage **1011** within the opening defined by lower plate **1012**, upper plate **1014**, left side plate **1016**, and right side plate **1018** of the cage **1011**. Further, the pluggable module **732** is in contact with protective covers **1026-1**, **1026-2**, **1026-3** (collectively, protective covers **1026**). Protective covers **1026**, in turn, are coupled to gap pads **1024-1**, **1024-2**, **1024-3**, respectively (collectively gap pads **1024**).

[0073] Insertion of pluggable module **1032** into receptacle **1010** results in a thermal connection being created between the receptacle **1010** and the pluggable module **1032**. When pluggable module **1032** is inserted into receptacle **1010**, each of the gap pads **1024** compresses by some amount. When pluggable module **1032** is inserted into receptacle **1010**, the pluggable module **1032** contacts the protective cover **1026** and pushes the protective cover **1026** away from a centerline of cage **1011**, i.e., in a generally radially outward direction. Specifically, the cover **1026-1** is pushed in the +x direction shown in FIG. **10**, cover **1026-2** is pushed in the +y direction shown in FIG. **10**, and cover **1026-3** is pushed in the -y direction. In FIG. **10**, the movement of cover **1026-1** relative to its resting position prior to insertion of the pluggable module **1032** is indicated by distance **d** (i.e., the pluggable module **1032** pushed the cover **1026-1** a distance **d** in the +x direction while being inserted). The movement of the covers **1026** away from the centerline of the cage **1011** compress the gap pads **1024** which are sandwiched between the covers **1026** and the cage **1011**. In other words, inserting the pluggable module **1032** into receptacle **1010** pushes against the protective cover **1026** such that the gap pad **1024** compresses during the insertion process.

[0074] The compression of the gap pads **1024** causes them to generate internal elastic restoring forces which urge the gap pads **1024** back towards their uncompressed state, similar to a spring

force generated by a spring under compression. These restoring forces push are applied to the protective covers **1026** in a direction opposite that of the compression—e.g., the restoring force of gap pad **1024-1** pushes cover **1026-1** in the $-x$ direction, the restoring force of gap pad **1024-2** pushes cover **1026-2** in the $-y$ direction, and the restoring force of gap pad **1024-3** pushes cover **1026-3** in the $+x$ direction. Consequently, the restoring force of a given gap pad **1024** is able to push the corresponding protective cover **1026** back against the pluggable module **1032**, which generates a contact pressure between the protective cover **1026** and the pluggable module **1032** maintains a contact therebetween. Said differently, the protective cover **1026** is pushed into full contact with the pluggable module **1032** by the restoring force of the gap pad **1024**, such that a thermal interface is created. Moreover, in some examples, protective covers **1032** may be sufficiently compliant/conformable that the restoring forces cause the protective covers **1026** to deform and comply to the outer surface of the pluggable module **1032**. This may allow for the number and size of air gaps at the interface between the pluggable module **1032** and the protective cover **1026** to be reduced, or in other words for the total area of mutual contact to be increased, which improves the rate of heat transfer across the thermal interface. In some examples, the restoring force from the gap pads **1024** is sufficiently strong to achieve a good thermal interface but not so strong as to require overly large insertion forces. For example, in some implementations a thermal interface between pluggable module and cage may transfer heat at a rate of at least 0.10 W/cm.sup.2C and the insertion forces required to insert the pluggable module **1032** into the receptacle **1010** may be no more than 25 lbf. In some examples, the pressure generated due to the restoring forces of the compressed gap pads **1024** may be between 5 lbf and 25 lbf when the gap pads **1024** are compressed. The gap pad **1024**, thus, serves the dual purpose of transferring heat from the protective covers **1026** to the cage **1011** and applying contact pressure between protective cover **1026** and pluggable module **1032** to assist the protective cover **1026** in creating a thermal connection between the receptacle **1010** and the pluggable module **1032**.

[0075] In some examples where gap pads **1024** and protective covers **1026** are in use, gap pads having different properties may be utilized. For example, different gap pads have different levels of thermal conductivity and different degrees of springiness and compressibility. Compressibility refers to the degree to which the gap pad compresses when subjected to a given amount of pressure and is inversely related to the degree to which the gap pad resists compression. That is, a highly compressible gap pad may lightly resist compression and thus may be compressed more for a given applied pressure, whereas a less compressible gap pad may more strongly resist compression and thus may be compressed less for a given applied pressure. Consequently, a gap pad that is more compressible will generate a lower restoring spring force when compressed by a given amount as compared to a less compressible gap pad compressed by the same amount. In other words, highly compressible gap pads tend to produce lower restoring forces and less compressible gap pads produce greater restoring forces, all other things being equal. It may be the case that a gap pad with a high degree of thermal conductivity has a higher degree of compressibility than a gap pad with a lower degree of thermal conductivity. Such a high thermal conductivity and high compressibility gap pad might generate a weak contact pressure between the protective cover **1026** and the pluggable module **1032**, which can result in a poor thermal interface between cover **1026** and module **1032**. Thus, the overall thermal conductivity of the cooling interface module **1022** may be poor, notwithstanding the high thermal conductivity of the gap pad. On the other hand, a gap pad with lower thermal conductivity may have higher compressibility and thus may generate a higher contact pressure between cover **1026** and module **1032**. Thus, such a low conductivity low compressibility gap pad may improve the thermal interface between the protective cover **1026** and module **1032**, but the low thermal conductivity of the gap pad itself may create a thermal bottleneck which reduces the overall thermal conductivity of the module **1022**. Thus, in some examples, to ensure that the cooling interface modules **1022** achieve a desired level of overall thermal conductivity, different gap pads having different degrees of thermal conductivity and

compressibility may be mixed together with the modules **1022**. For example, a given module **1022** may be provided with multiple gap pads **1024** attached to the same protective cover **1026**, including one or more gap pads **1024** with high thermal conductivity and high compressibility to ensure highly conductive thermal pathway between cover **1026** and cage **1011**, and one or more gap pads **1024** with lower thermal conductivity and lower compressibility to ensure sufficient contact pressure is applied between protective cover **1026** and module **1032**. These different gap pads **1024** may be arranged in various patterns on the protective cover **1026**, such as a pattern with the highly conductive gap pad(s) **1024** arranged in a middle region and the less conductive gap pad(s) **1024** surrounding the highly conductive gap pads, or a pattern in which highly conductive gap pads **1024** and less conductive gap pads **1024** are arranged in alternating positions of an array (such as a checkerboard pattern).

[0076] In some other examples, a first gap pad with a high degree of thermal conductivity may be installed adjacent to one plate. Additional gap pads having a lower degree of thermal conductivity but greater compressibility may be installed adjacent to other plates. Because each gap pad is compressed when a pluggable module **1032** is inserted into receptacle **1010**, the gap pads having greater compressibility will generate a lower restoring force than the less compressible gap pads.

[0077] In addition, in some examples, additional compressible materials may be coupled to the gap pad **1024** and/or the protective cover **1026**. In such examples, the additional compressible material may be used to increase the amount of restoring force pushing the protective cover against the pluggable module **1032**. This may aid in creating a good thermal interface between the pluggable module **732** and receptacle **1010**.

[0078] FIG. **11** is an example method **80** for use of a cooling interface for a pluggable module consistent with the present disclosure. At **82**, method **80** may include inserting a removably pluggable module into a receptacle of a computing system. As described previously, the receptacle may be designed to receive a removably pluggable module within a bay defined by a plurality of walls. The receptacle may be coupled with the computing system or may be integrally formed as part of a component of the computing system, such as formed as part of a chassis.

[0079] At **84**, method **80** may include contacting the removably pluggable module with a protective cover of a cooling interface module attached to the receptacle. As described previously-, the receptacle may include at least one cooling interface module coupled to a plate of the receptacle. The cooling interface module may comprise a protective cover, such as a thin metal sheet, and a gap pad, and may be coupled to the receptacle such that the gap pad is sandwiched between the receptacle and the protective cover. Thus, when the removably pluggable module is inserted into the receptacle, the removably pluggable module may be placed in contact with the protective cover.

[0080] At **86**, method **80** may include compressing a gap pad of the cooling interface module. As described previously, the gap pad may have an inherent compressibility, such that when the removably pluggable module is inserted into the receptacle, the gap pad is compressed to allow the insertion. Said differently, when the removably pluggable module contacts the protective cover during insertion into the receptacle, the gap pad may also compress to allow the removably pluggable module to be fully inserted into and received by the receptacle.

[0081] At **88**, method **80** may include thermally coupling the removably pluggable module to the receptacle by causing the gap pad to push the protective cover against the removably pluggable module. As described with respect to FIG. **10**, because the gap pad undergoes compression when the removably pluggable module is inserted into the cage, a restoring force moves the gap pad back towards its uncompressed state once the removably pluggable module is fully inserted. Because the gap pad is attached to the protective cover, the restoring force of the gap pad has the effect of also pushing the protective cover against the removably pluggable module. As described previously, particularly with respect to FIG. **1**, the protective cover is made of a thermally conductive material and, since the gap pad is also thermally conductive, the contact between the outer portion of the receptacle, the gap pad, the protective cover, and the removably pluggable module results in a

thermal connection between the removably pluggable module and the receptacle.

[0082] The receptacle may be thermally coupled to a cold plate of the computing system. As discussed with respect to FIGS. 2, 8A-8B and 9A-9B, the cold plate may be either directly coupled to the receptacle or may be remotely located such the receptacle is thermally coupled to the cold plate by, for example, heat pipes. However, in either configuration, the thermal coupling of the removably pluggable module to the receptacle results in the removably pluggable module being thermally coupled to the cold plate. In this manner, the removably pluggable module is able to be cooled.

[0083] It is to be understood that both the general description and the detailed description provide example implementations that are explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. Other examples in accordance with the present disclosure will be apparent to those skilled in the art based on consideration of the disclosure herein. For example, various mechanical, compositional, structural, electronic, and operational changes may be made to the disclosed examples without departing from the scope of this disclosure, including for example the addition, removal, alteration, substitution, or rearrangement of elements of the disclosed examples, as would be apparent to one skilled in the art in consideration of the present disclosure. Moreover, it will be apparent to those skilled in the art that certain features or aspects of the present teachings may be utilized independently (even if they are disclosed together in some examples) or may be utilized together (even if disclosed in separate examples), whenever practical. In some instances, well-known circuits, structures, and techniques have not been shown or described in detail in order not to obscure the examples. Thus, the following claims are intended to be given their fullest breadth, including equivalents, under the applicable law, without being limited to the examples disclosed herein.

[0084] References herein to examples, implementations, or other similar references should be understood as referring to prophetic or hypothetical examples, rather than to devices/systems that have been actually produced, unless explicitly indicated otherwise. Similarly, references to qualities or characteristics of examples should be understood as representing the educated estimates or expectations of the inventors based on their understanding of the relevant principles involved, application of theory and/or modeling, and/or past experiences, rather than as being representations of the actual qualities or characteristics of an actually produced device/system or the empirical results of tests actually carried out, unless explicitly indicated otherwise.

[0085] Further, spatial, positional, and relational terminology used herein is chosen to aid the reader in understanding examples of the invention but is not intended to limit the invention to a particular reference frame, orientation, or positional relationship. For example, spatial, positional, and relational terms such as “up”, “down”, “lateral”, “beneath”, “below”, “lower”, “above”, “upper”, “proximal”, “distal”, and the like may be used herein to describe directions or to describe one element's or feature's spatial relationship to another element or feature as illustrated in the figures. These spatial terms are used relative to reference frames in the figures and are not limited to a particular reference frame in the real world. Furthermore, if a different reference frame is considered than the one illustrated in the figures, then the spatial terms used herein may need to be interpreted differently in that different reference frame. Moreover, the poses of items illustrated in the figure are chosen for convenience of illustration and description, but in an implementation in practice the items may be posed differently.

[0086] In addition, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. Moreover, the terms “comprises”, “comprising”, “includes”, and the like specify the presence of stated features, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups. Components described as coupled may be electronically or mechanically directly coupled, or they may be indirectly coupled via one or more

intermediate components, unless specifically noted otherwise.

[0087] And/or: Occasionally the phrase “and/or” is used herein in conjunction with a list of items. This phrase means that any combination of items in the list—from a single item to all of the items and any permutation in between—may be included. Thus, for example, “A, B, and/or C” means “one of {A}, {B}, {C}, {A, B}, {A, C}, {C, B}, and {A, C, B}”.

[0088] Mathematical and geometric terms are not necessarily intended to be used in accordance with their strict definitions unless the context of the description indicates otherwise, because a person having ordinary skill in the art would understand that, for example, a substantially similar element that functions in a substantially similar way could easily fall within the scope of a descriptive term even though the term also has a strict definition. Moreover, unless otherwise noted herein or implied by the context, when terms of approximation such as “substantially,” “approximately,” “about,” “around,” “roughly,” and the like, are used, this should be understood as meaning that mathematical exactitude is not required and that instead a range of variation is being referred to that includes but is not strictly limited to the stated value, property, or relationship. In particular, in addition to any ranges explicitly stated herein (if any), the range of variation implied by the usage of such a term of approximation includes at least any inconsequential variations and also those variations that are typical in the relevant art for the type of item in question due to manufacturing or other tolerances. In any case, the range of variation may include at least values that are within +1% of the stated value, property, or relationship unless indicated otherwise.

Claims

1. A receptacle for receiving a pluggable module, comprising: a cage comprising a plurality of plates including: a lower plate; an upper plate; a left side plate disposed between the lower plate and the upper plate; a right side plate disposed between the lower plate and the upper plate opposite the left side plate; a bay defined by the plurality of plates, wherein the bay is sized to receive a pluggable module; one or more cooling interface modules, wherein: each cooling interface module of the one or more cooling interface modules comprises: a gap pad; and a protective cover; and each cooling interface module of the one or more cooling interface modules is disposed such that the gap pad is adjacent to a given plate of the plurality of plates and located between the protective cover and the given plate.
2. The receptacle of claim 1, wherein the protective cover is movable with respect to the given plate.
- 3 The receptacle of claim 1, wherein each cooling interface module of the one or more cooling interface modules further comprises a hook attached to one end of the protective cover and configured to pivotably connect the cooling interface module to an edge of the cage.
4. The receptacle of claim 1, wherein: a first cooling interface module of the one or more cooling interface modules is disposed adjacent to a first plate of the plurality of plates; and a second cooling interface module of the one or more cooling interface modules is disposed adjacent to a second plate of the plurality of plates.
5. The receptacle of claim 4, wherein a third cooling interface module of the one or more cooling interface modules is disposed adjacent to a third plate of the plurality of plates.
6. The receptacle of claim 1, wherein the protective cover comprises a metal sheet.
7. The receptacle of claim 6, wherein: the metal sheet has an average thickness of 0.003 inches or less; and the metal sheet comprises one of copper, a copper alloy, nickel, steel, stainless steel, aluminum, an aluminum alloy, or any combination of these.
8. The receptacle of claim 1, wherein the protective cover comprises a low-friction, conforming, tear resistant, and thermally conductive material.
9. A computing system, comprising: a chassis; a system board supported by the chassis; a removably pluggable module; a receptacle coupled to or formed by the chassis and configured to receive the removably pluggable module, wherein the receptacle comprises: a cage comprising a

plurality of plates including: a lower plate; an upper plate located opposite the lower plate; and side plates coupled to the lower plate and to the upper plate; at least one of cooling interface module disposed along a plate of the plurality of plates, wherein the at least one cooling interface module further comprises: a gap pad; and a protective cover coupled to the gap pad such that the gap pad is sandwiched between a plate of the cage and the protective cover; and a cold plate thermally coupled to the receptacle.

10. The system of claim 9, wherein the gap pad of at least one cooling interface module compresses upon insertion of the removably pluggable module such that a thermal connection is created between the receptacle and the removably pluggable module.

11. The system of claim 10, wherein the removably pluggable module compresses the gap pad by pushing the protective cover away from the removably pluggable module as the removably pluggable module is inserted into the receptacle.

12. The system of claim 11, wherein contact between the gap pad and the receptacle, between the gap pad and the protective cover, and between the protective cover and the pluggable module creates a thermal interface between the pluggable module and the receptacle.

13. The system of claim 10, wherein: insertion of the removably pluggable module results in compression of the gap pad; and the protective cover is pushed against the removably pluggable module by a restoring force caused by compression of the gap pad.

14. The system of claim 9, wherein the gap pad has a compression rate of between 10 and 50 percent.

15. The system of claim 14, wherein the gap pad has a compression rate of twenty percent.

16. The system of claim 9, wherein the cold plate is directly coupled to the receptacle.

17. The system of claim 9, wherein: the cold plate is a remotely located cold plate; and the receptacle is thermally coupled to the remotely located cold plate by one or more thermal transfer devices.

18. The system of claim 9, wherein the protective covering is a stainless steel covering.

19. A method comprising: inserting a removably pluggable module into a receptacle of a computing system, the receptacle comprising a cage defining a bay into which the removably pluggable module is inserted and a cooling interface module attached to the; during the inserting, contacting the removably pluggable module with a protective cover of the cooling interface module; by the contacting, compressing a gap pad of the cooling interface module between the protective cover and a plate of the cage, wherein the gap pad is attached to the protective cover; and thermally coupling the removably pluggable module to the receptacle by causing the gap pad to, in response to being compressed, push the protective cover against the removably pluggable module.

20. The method of claim 18, wherein: the receptacle is thermally coupled to a cold plate of the computing system; and thermally coupling the removably pluggable module to the receptacle comprises thermally coupling the removably pluggable module to the cold plate.
